

Dynamic Guideline: WHO-Based Maternal Health - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note: Maternal health is not a single disease entity; this report synthesizes WHO guidance across the continuum of pregnancy, childbirth, postpartum care, maternal infection prevention/treatment, and postpartum haemorrhage. Exactly **5 WHO maternal-health clinical guidelines** are synthesized below.

Required title check: The title “**Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline**” appears to describe a guideline-development case study rather than a WHO clinical guideline for maternal diagnosis, treatment, or management. It is therefore not counted among the 5 clinical WHO guidelines below.

#	WHO Guideline Title	Publication / Update Date	Target Population	Scope and Objectives
1	WHO recommendations on antenatal care for a positive pregnancy experience	2016	Pregnant women and adolescent girls; fetuses; health-care providers and health systems	Routine antenatal care, risk detection, nutrition, infection screening, fetal assessment, health-system contacts, and respectful care.
2	WHO recommendations: intrapartum care for a positive childbirth experience	2018	Women and adolescent girls in labour; newborns; maternity-care providers	Evidence-based care during labour and childbirth, including respectful care, labour monitoring, pain relief, interventions, and avoidance of unnecessary procedures.
3	WHO recommendations for the prevention and treatment of maternal peripartum infections	2015	Pregnant, intrapartum, and postpartum women; maternity-care providers	Prevention and treatment of infection around childbirth, including antibiotic prophylaxis, diagnosis, and treatment of puerperal infection/sepsis.

4 WHO recommendations on maternal and newborn care for a positive postnatal experience	2022 Postpartum women, newborns, families, and health systems	Postnatal contacts, maternal assessment, breastfeeding support, mental health, contraception, newborn care, and health-system continuity.
5 Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage	2025 Pregnant and postpartum women; birth attendants; emergency obstetric teams; health systems	Prevention, early detection, diagnosis, and emergency treatment of postpartum haemorrhage, including uterotonics, tranexamic acid, bundles of care, referral, blood products, and surgical escalation.

Most recent clinical WHO guideline in this maternal-health set:
Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage.

1.2 Key Recommendations

GRADE Interpretation Used

GRADE Category	Meaning
High	Further research is very unlikely to change confidence in the effect estimate.
Moderate	Further research may affect confidence and may change the estimate.
Low	Further research is likely to affect confidence and may change the estimate.
Very Low	Effect estimate is very uncertain.
Strong Recommendation	Benefits clearly outweigh harms/burden, or vice versa; applies to most patients.
Conditional Recommendation	Benefits probably outweigh harms/burden, but values, context, feasibility, or evidence certainty vary.

A. Antenatal Care

Recommendation	GRADE Evidence Quality	Strength
Provide antenatal care using a model with a minimum of 8 contacts to reduce perinatal mortality and improve women's experience of care.	Moderate	Strong
Provide daily oral iron and folic acid supplementation during pregnancy, typically 30–60 mg elemental iron plus 400 µg folic acid, according to anaemia prevalence and individual risk.	Moderate	Strong

Provide calcium supplementation , usually 1.5–2.0 g elemental calcium daily, in populations with low dietary calcium intake to reduce risk of pre-eclampsia.	Moderate	Strong
Offer early ultrasound before 24 weeks' gestation where available to estimate gestational age, detect multiple pregnancy, and improve fetal anomaly detection.	Moderate	Conditional
Screen for and manage common pregnancy risks, including hypertension, anaemia, infections, diabetes risk, fetal growth concerns, intimate partner violence, tobacco/alcohol use, and mental health symptoms.	Low– Moderate	Strong
Provide respectful, person-centred antenatal care, including privacy, informed choice, culturally appropriate communication, and adolescent-responsive services.	Low	Strong

B. Intrapartum Care

Recommendation	GRADE Evidence Quality	Strength
Provide respectful maternity care for all women during labour and childbirth, preserving dignity, privacy, confidentiality, informed consent, and freedom from mistreatment.	Low– Moderate	Strong
Allow a companion of choice during labour and childbirth.	Moderate	Strong
Do not use routine admission cardiotocography for low-risk women in spontaneous labour where intermittent auscultation is appropriate.	Low– Moderate	Conditional
Do not use routine amniotomy, routine oxytocin augmentation, or fixed labour-duration thresholds to accelerate normally progressing labour.	Low– Moderate	Conditional
Encourage mobility, oral fluids, choice of birth position, and non-pharmacologic or pharmacologic pain relief according to the woman's preferences and clinical context.	Low– Moderate	Strong
Avoid routine episiotomy ; use restrictive episiotomy only when clinically indicated.	Moderate	Strong
Use active management of the third stage of labour, especially administration of a uterotonic, to reduce postpartum haemorrhage.	High	Strong

C. Prevention and Treatment of Maternal Peripartum Infections

Recommendation	GRADE Evidence Quality	Strength
Give prophylactic antibiotics before caesarean section , preferably before skin incision, to reduce maternal infectious morbidity.	High	Strong

Do not give routine antibiotic prophylaxis for uncomplicated vaginal birth.	Low– Moderate	Strong
Give antibiotic prophylaxis for selected high-risk obstetric procedures, such as manual removal of placenta, according to local protocols and antimicrobial-resistance patterns.	Low	Conditional
Suspect maternal sepsis in postpartum women with fever, tachycardia, hypotension, uterine tenderness, foul-smelling lochia, altered mental status, or organ dysfunction.	Low	Strong
Treat suspected puerperal sepsis urgently with broad-spectrum antibiotics, source control when needed, fluid resuscitation, monitoring, and escalation to higher-level care.	Moderate	Strong
Maintain infection-prevention measures during childbirth, including hand hygiene, sterile technique for procedures, clean birth environment, and rational antibiotic use.	Moderate	Strong

D. Postnatal Maternal and Newborn Care

Recommendation	GRADE Evidence Quality	Strength
Ensure women and newborns receive postnatal care within the first 24 hours after birth , with continued scheduled contacts during the first 6 weeks.	Moderate	Strong
Assess postpartum women for bleeding, uterine involution, infection, pain, blood pressure, urinary symptoms, breastfeeding difficulties, mood disorders, and danger signs.	Low– Moderate	Strong
Support early and exclusive breastfeeding unless medically contraindicated, including skilled lactation support where needed.	Moderate	Strong
Screen for postpartum depression and anxiety where systems exist for diagnosis, treatment, follow-up, and referral.	Low– Moderate	Conditional
Provide counselling and access to postpartum contraception, birth spacing, sexual health, and return-to-care planning.	Moderate	Strong
Educate women and families on danger signs: heavy bleeding, fever, severe headache, visual symptoms, chest pain, dyspnoea, seizures, severe abdominal pain, calf swelling, or thoughts of self-harm.	Low	Strong

E. Postpartum Haemorrhage: Prevention, Diagnosis, and Treatment

Recommendation	GRADE Evidence Quality	Strength
Give a uterotonic to all women during the third stage of labour to prevent postpartum haemorrhage.	High	Strong

Use oxytocin 10 IU IM/IV as the preferred uterotonic where quality-assured oxytocin and cold-chain storage are available.	High	Strong
Where oxytocin is unavailable or quality cannot be assured, use an alternative uterotonic such as heat-stable carbetocin, misoprostol, ergometrine/methylergometrine, or oxytocin–ergometrine, considering contraindications.	Moderate	Conditional
Use objective blood-loss assessment where feasible, combined with vital signs and clinical assessment, to improve early recognition of postpartum haemorrhage.	Moderate	Strong
For diagnosed postpartum haemorrhage, use a first-response treatment bundle including uterine massage, uterotonics, tranexamic acid, IV fluids, examination for cause, and timely escalation.	High	Strong
Give tranexamic acid 1 g IV as soon as possible and within 3 hours of birth for women with clinically diagnosed postpartum haemorrhage; repeat once if bleeding continues after 30 minutes or restarts within 24 hours.	High	Strong
Use uterine balloon tamponade only where appropriate emergency obstetric care, monitoring, blood transfusion, and surgical backup are available.	Low– Moderate	Conditional
Escalate promptly to blood products, uterine tamponade, compression sutures, artery ligation, embolization, or hysterectomy when bleeding persists despite first-response measures.	Low– Moderate	Strong

1.3 Guideline Development Methodology

WHO guideline development generally follows the WHO Handbook for Guideline Development and includes:

Evidence Review Process

- Formulation of clinical questions using **PICO**: population, intervention, comparator, outcome.
- Systematic reviews and evidence-to-decision frameworks.
- Assessment of:
 - Mortality and severe morbidity.
 - Maternal experience and respectful care.
 - Adverse effects.
 - Feasibility.
 - Equity.
 - Resource implications.
 - Acceptability to women, families, and providers.
- GRADE assessment of certainty of evidence.

Consensus Method

- Multidisciplinary Guideline Development Groups.
- Representation from obstetrics, midwifery, public health, epidemiology, neonatology, pharmacology, implementation science, and patient/community perspectives.
- Structured discussion of evidence-to-decision criteria.
- Formal agreement on direction and strength of recommendations.

Conflict of Interest Management

- Declaration of financial, academic, and institutional conflicts.
 - WHO review and management of conflicts before participation.
 - Exclusion or restricted participation where conflicts may compromise independence.
 - Documentation of decisions in guideline annexes or methodology sections.
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II. Evidence Summary After WHO Latest Guideline Release (Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence considered for this update focused on maternal health after release of the latest WHO postpartum haemorrhage guideline, especially:

- Postpartum haemorrhage prevention.
- Early detection and diagnosis of postpartum haemorrhage.
- First-response treatment bundles.
- Tranexamic acid.
- Uterotonics.
- Uterine balloon tamponade.
- Maternal sepsis.
- Antenatal and postnatal care interventions affecting maternal morbidity and mortality.

Databases and Sources Searched

- WHO official guideline and technical-document repositories.
- PubMed/MEDLINE.
- Cochrane Database of Systematic Reviews.
- ClinicalTrials.gov.
- WHO International Clinical Trials Registry Platform.
- Major journals: **The Lancet**, **New England Journal of Medicine**, **JAMA**, **BMJ**, **BJOG**, **Obstetrics & Gynecology**.
- Authoritative public-health bodies: CDC, NIH, NICE/NHS, FIGO, and professional obstetric societies where relevant.

Inclusion Criteria

- Randomized controlled trials, cluster randomized trials, systematic reviews, meta-analyses, implementation studies, and major observational studies.
- Maternal health interventions relevant to WHO guideline recommendations.
- Studies reporting clinically important outcomes: maternal death, severe maternal morbidity, severe postpartum haemorrhage, transfusion, hysterectomy, sepsis, thromboembolism, breastfeeding outcomes, postpartum depression, or health-system access.
- Studies applicable to low-, middle-, or high-income settings.

Exclusion Criteria

- Non-systematic narrative reviews without new data.
- Case reports unless reporting rare but serious harms.
- Studies not involving pregnant, intrapartum, or postpartum populations.
- Studies without clinically relevant maternal outcomes.
- Duplicates or superseded analyses.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
E-MOTIVE Trial Collaborators. Early detection and treatment bundle for postpartum haemorrhage. NEJM, 2023.	Cluster randomized trial	A strategy combining calibrated blood-loss measurement and a first-response treatment bundle reduced severe postpartum haemorrhage, laparotomy, and maternal death composite outcomes compared with usual care. Supports WHO emphasis on early detection and bundled care.	High
WOMAN Trial Collaborators. Tranexamic acid for treatment of postpartum haemorrhage. Lancet, 2017.	Large international randomized placebo-controlled trial	Early IV tranexamic acid reduced death due to bleeding, especially when given within 3 hours of birth, without clear increase in thromboembolic events. Remains foundational evidence for WHO TXA recommendation.	High
WOMAN-2 Trial Collaborators. Tranexamic acid for prevention of postpartum bleeding in women with anaemia. Lancet, 2024.	Randomized placebo-controlled trial	Prophylactic TXA in anaemic women did not clearly reduce clinically diagnosed postpartum haemorrhage overall, although some severe bleeding outcomes may have been reduced. Does not justify routine prophylactic TXA for all anaemic women outside guideline-supported contexts.	Moderate

Widmer et al. Heat-stable carbetocin versus oxytocin to prevent postpartum haemorrhage. NEJM, 2018.	Randomized non-inferiority trial	Heat-stable carbetocin was non-inferior to oxytocin for prevention of postpartum haemorrhage after vaginal birth and may be useful where oxytocin cold-chain reliability is limited.	High
Cochrane and network meta-analyses of uterotonics for prevention of postpartum haemorrhage.	Systematic review / network meta-analysis	Oxytocin remains first-line where quality assured; carbetocin, misoprostol, ergometrine, and oxytocin–ergometrine are effective alternatives but differ in adverse-effect profiles and feasibility.	High
Evidence on uterine balloon tamponade in low-resource settings.	Trials and systematic reviews	Balloon tamponade may control refractory atonic bleeding in selected settings but can be harmful if used where surgical care, blood transfusion, monitoring, and escalation are unavailable. Supports conditional WHO recommendation.	Low–Moderate
Postnatal care implementation studies.	Observational and implementation research	Early postnatal contact improves detection of bleeding, hypertension, infection, breastfeeding problems, and mental-health symptoms, but evidence varies by system capacity.	Low–Moderate
Maternal sepsis recognition and treatment evidence.	Observational studies, sepsis bundles, guideline extrapolation	Early recognition, prompt antibiotics, fluids, source control, and escalation are associated with improved outcomes, although pregnancy-specific RCT evidence is limited.	Low–Moderate

2.3 Evidence Gaps and Future Research Needs

Postpartum Haemorrhage

- Optimal implementation of objective blood-loss measurement in low-resource and high-volume birth settings.
- Best design of first-response postpartum haemorrhage bundles for different health-system levels.
- Comparative effectiveness of uterotonics when cold-chain integrity is unreliable.
- Role of prophylactic tranexamic acid in specific high-risk groups, such as severe anaemia, caesarean birth, placenta accreta spectrum, or previous postpartum haemorrhage.
- Safer criteria for uterine balloon tamponade use in facilities without immediate surgical backup.
- Blood-product protocols adapted to low-resource settings.

Antenatal Care

- Best models for achieving 8 effective contacts in rural, conflict-affected, or under-resourced settings.
- Integration of digital antenatal care without worsening inequity.
- Optimal screening approaches for pre-eclampsia, fetal growth restriction, gestational diabetes, mental health disorders, and intimate partner violence.
- Implementation of adolescent-responsive antenatal care.

Intrapartum Care

- Reduction of unnecessary caesarean section without compromising maternal or neonatal outcomes.
- Respectful maternity care interventions that sustainably reduce abuse, coercion, and neglect.
- Locally validated labour-monitoring tools replacing outdated fixed labour-duration thresholds.

Postnatal Care

- Effective models for early postnatal home visits.
- Postpartum hypertension surveillance and treatment pathways.
- Integration of mental-health screening with treatment access.
- Postpartum contraception uptake while protecting informed choice.

Maternal Infection and Sepsis

- Pregnancy-specific sepsis screening tools.
- Antibiotic stewardship in obstetric care.
- Optimal prophylaxis for operative vaginal birth and severe perineal trauma in different resistance settings.
- Sepsis bundle implementation in district hospitals and primary maternity units.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Use oxytocin 10 IU IM/IV as first-line uterotonic for prevention of postpartum haemorrhage.	Strong evidence continues to support oxytocin; heat-stable carbetocin is a feasible alternative where cold-chain problems exist.	Maintain. Use quality-assured oxytocin first-line; consider heat-stable carbetocin or other WHO-listed uterotonics where oxytocin quality cannot be assured.
Use objective blood-loss assessment to improve early diagnosis of postpartum haemorrhage.	E-MOTIVE supports calibrated blood-loss measurement combined with rapid bundled treatment.	Strengthen implementation priority. Use calibrated drapes or standardized quantitative blood-loss tools where feasible, with vital-sign assessment and immediate response triggers.

Treat postpartum haemorrhage with a first-response bundle.	Cluster-trial evidence supports improved outcomes with bundled early care.	Maintain and operationalize. Facilities should implement a standardized first-response bundle: uterine massage, uterotonic, TXA, IV fluids, examination for cause, call for help, and escalation.
Give tranexamic acid for treatment of diagnosed postpartum haemorrhage within 3 hours of birth.	WOMAN trial evidence remains high certainty; later prophylaxis evidence does not replace treatment indication.	Maintain. Give TXA 1 g IV immediately once PPH is diagnosed and within 3 hours of birth; repeat once if bleeding continues or restarts.
Use prophylactic TXA routinely to prevent postpartum haemorrhage.	WOMAN-2 does not support broad routine prophylactic TXA for anaemic women; evidence remains context-specific.	Do not introduce routine universal prophylactic TXA. Consider only in specific protocols, trials, or high-risk contexts where supported by local policy.
Use uterine balloon tamponade for refractory postpartum haemorrhage.	Evidence suggests benefit depends heavily on facility readiness and ability to escalate.	Maintain conditional recommendation. Use only where monitoring, blood transfusion, emergency surgery, and referral pathways are available.
Provide early postnatal care within 24 hours and continued contacts.	Implementation evidence supports earlier detection of bleeding, hypertension, sepsis, breastfeeding problems, and mental-health concerns.	Maintain and prioritize. Ensure first 24-hour assessment and scheduled postpartum contacts, with clear danger-sign counselling.
Give prophylactic antibiotics before caesarean section.	Evidence remains high certainty; timing before incision improves prevention of infection.	Maintain. Administer prophylactic antibiotics before skin incision unless contraindicated, following local antimicrobial-resistance guidance.

References

1. World Health Organization. **WHO recommendations on antenatal care for a positive pregnancy experience.** Geneva: WHO; 2016.
2. World Health Organization. **WHO recommendations: intrapartum care for a positive childbirth experience.** Geneva: WHO; 2018.
3. World Health Organization. **WHO recommendations for the prevention and treatment of maternal peripartum infections.** Geneva: WHO; 2015.
4. World Health Organization. **WHO recommendations on maternal and newborn care for a positive postnatal experience.** Geneva: WHO; 2022.
5. World Health Organization. **Consolidated guidelines for the prevention, diagnosis and treatment of postpartum haemorrhage.** Geneva: WHO; 2025.
6. WOMAN Trial Collaborators. **Effect of early tranexamic acid administration on mortality, hysterectomy, and other morbidities in women with postpartum haemorrhage.** *The Lancet*. 2017;389:2105–2116.
7. E-MOTIVE Trial Collaborators. **Early detection and treatment of postpartum**

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8. Widmer M, et al. **Heat-stable carbetocin versus oxytocin to prevent hemorrhage after vaginal birth.** *New England Journal of Medicine*. 2018;379:743–752.
 9. Gallos ID, et al. **Uterotonic agents for preventing postpartum haemorrhage: a network meta-analysis.** *Cochrane Database of Systematic Reviews / associated evidence synthesis*. 2018.
 10. WOMAN-2 Trial Collaborators. **Tranexamic acid for the prevention of postpartum bleeding in women with anaemia.** *The Lancet*. 2024.